Department of Physics, Princeton University

Graduate Preliminary Examination
Part I

Thursday, January 9, 2020
9:00 am - 12:00 noon

Answer TWO out of the THREE questions in Section A (Mechanics) and TWO out of the THREE questions in Section B (Electricity and Magnetism).

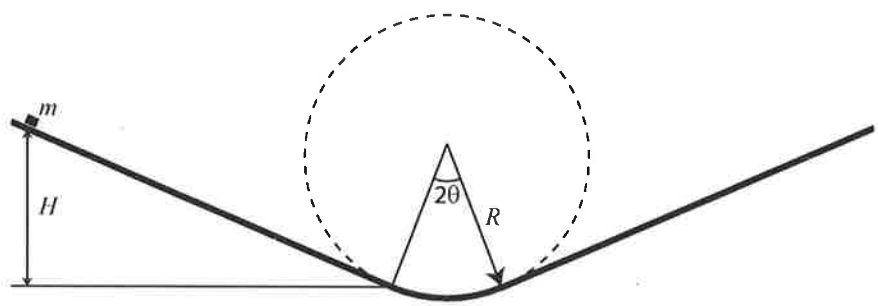
Work each problem in a separate examination booklet.

Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Mechanics

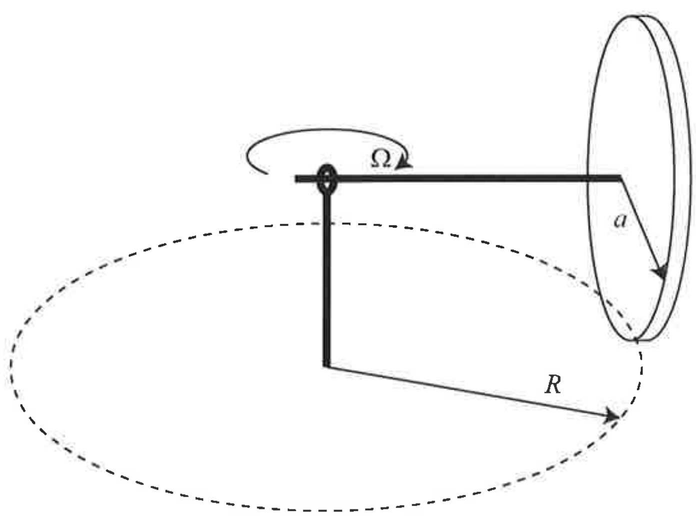
1. **Inclined track**

   A small block of mass m is sliding without friction on a track consisting of two straight surfaces inclined at angles $\pm\theta$ and a circular arc of radius R, as shown in the figure. The block starts at a height H above the transition point from straight to circular path. Find the period of oscillations, i.e, the time it takes for the block to return to its starting point. You can assume that angle θ is small.



2. **Rolling disk**

A thin solid disk of mass M and radius a is rolling without slipping in a circle of radius R. The disk is attached to a horizontal axle, which is attached to a vertical shaft at the center of the circle with a joint that can apply forces but no torques. The horizontal axle and the disk rotate around the vertical axis with angular frequency Ω as the disk rolls on the ground. The plane of the disk always remains vertical.



(a) Find the normal force F_N that the disk exerts on the ground.

(b) Suppose now that the disk experiences rolling friction with the ground with a coefficient of rolling friction μ. There is no friction in other joints and there are no external driving forces. Find the differential equation that describes how the angular frequency Ω will decay with time.

3. **Collecting dust**

A solid planet of mass M and radius R is moving with velocity v through a dilute dust cloud with a mass density per unit volume ρ. The particles in the dust cloud are sufficiently dilute that you can ignore their collisions or gravitational self-interactions. Calculate the rate dM/dt at which the planet will accrete mass from the dust cloud.

Section B. Electricity and Magnetism

1. **Rotating charged sphere**

   Electromagnetic radiation is observed to originate from a system consisting of an electrically charged sphere of radius R placed in a uniform magnetic field B and spinning about its axis with a large angular velocity ω. The total charge Q on the sphere is uniformly distributed over its volume and fixed in place. The sphere has mass M with a constant mass density. The spin axis of the sphere, which is free to move, makes an angle α with the direction of the uniform B field. Assume R is small compared to the wavelength of the radiation.

   (a) Calculate the magnitude and direction of the magnetic moment of the rotating sphere.

   (b) Treating the sphere as a gyroscope, calculate the precession frequency Ω of the spin axis.

   (c) Find the time-averaged radiated power from this system.

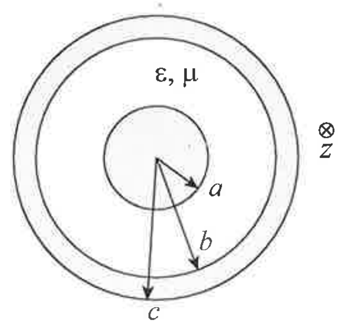
2. **Current between conductors in a liquid**

   (a) Consider two perfect conductors of arbitrary shape immersed in a large tank of weakly conducting liquid with electrical conductivity σ. Show that the electrical resistance between conductors can be related to their capacitance. For definiteness, assume that the two conductors are held at potentials $+V$ and $-V$ and the walls of the large tank containing the liquid are grounded.

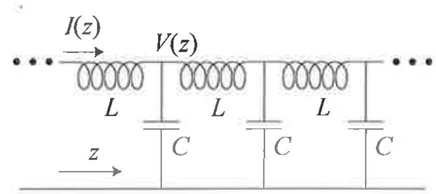
   (b) Consider two conducting spheres with equal radius a immersed in the liquid. The centers of the spheres are separated by a distance $b >> a$. Find the electrical resistance between the spheres.

3. **Transmission line**

Consider a transmission line consisting of two perfect coaxial cylindrical conductors extending indefinitely along z direction. The space between the conductors is filled with a material that has electric permeability ϵ and magnetic susceptibility μ.



Such two-conductor transmission line supports TEM waves propagating along z. The transmission line can also be represented by an equivalent circuit shown below, where L and C are the inductance and capacitance per unit length.



(a) Using the equivalent circuit, find differential equations for the voltage $V(z)$ and the current $I(z)$ along the transmission line in terms of the inductance and capacitance per unit length.

(b) Using Maxwell equations, write down a general form of the propagating TEM wave. Find the speed of propagation as well as the relationship between E and B fields in the space between the conductors. What are the E and B fields inside the conductors?

(c) Calculate the capacitance and the inductance per unit length for the transmission line with dimensions shown in the figure. Show that the product LC reduces to a simple result independent of the dimensions of the conductors.

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Graduate Preliminary Examination
Part II

Friday, January 10, 2020
9:00 am - 12:00 noon

Answer TWO out of the THREE questions in Section A (Quantum Mechanics) and TWO out of the THREE questions in Section B (Thermodynamics and Statistical Mechanics).

Work each problem in a separate booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Quantum Mechanics

1. **Low-energy scattering.**

   Consider scattering of a quantum mechanical particle of mass m from an *attractive* spherical delta-function shell potential

   $$V(\vec{r}) = -\alpha\delta(|\vec{r}| - a),$$

   where $\alpha > 0$ and $a > 0$ are constants. Consider the low energy limit where the momentum $\hbar k$ of the incoming particle obeys $ka \ll 1$.

   (a) For most values of a (keeping α fixed) the scattering cross-section can be approximated by a simple expression that is independent of k for $ka \ll 1$. However in the vicinity of a particular value $a = a_0$ its behavior is more complicated. Calculate the total scattering cross-section away from a_0.

   (b) What is the total scattering cross-section if $a = a_0$?

   (c) Consider now rotationally-invariant bound states of the particle in the potential V. Show the potential either has zero or one such bound states. Find the critical value of a (for a given α) where the bound state first appears.

2. **Degeneracies of the 2D harmonic oscillator.**

A spinless particle of mass m is placed into a 2D harmonic oscillator potential with classical frequency ω:

$$H = \frac{\hbar\omega}{2}\left(P_x^2 + P_y^2 + X^2 + Y^2\right) = \frac{\hbar\omega}{2}\left[-\frac{1}{R}\frac{\partial}{\partial R}R\frac{\partial}{\partial R} - \frac{1}{R^2}\frac{\partial^2}{\partial\phi^2} + R^2\right],$$

where, in order to avoid clutter, we define the dimensionless operators:

$$X = x\sqrt{m\omega/\hbar}\,, \qquad Y = y\sqrt{m\omega/\hbar}\,, \qquad P_x = p_x/\sqrt{\hbar m\omega}\,, \qquad P_y = p_y/\sqrt{\hbar m\omega}\,,$$

as well as the polar coordinates (R, ϕ) obeying $X = R\cos\phi$ and $Y = R\sin\phi$.

(a) What are the energy levels of H and their degeneracies?

(b) What is the unnormalized wavefunction $\psi_0(R, \phi)$ for the ground state?

(c) The potential is rotationally symmetric, so the angular momentum

$$L = \hbar(XP_y - YP_x)$$

is conserved. However, the conservation of L by itself cannot explain the presence of all degeneracies in the spectrum of H, so this suggests that there are more conserved quantities. Indeed, here are two more:

$$J_1 = (\hbar/4)(X^2 - Y^2 + P_x^2 - P_y^2)\,, \qquad J_2 = (\hbar/2)(XY + P_xP_y)\,.$$

It is straightforward to check that both J_1 and J_2 commute with H. Interestingly, J_1, J_2, and L have the same commutation relations with each other as the 3D angular momentum operators! In the usual normalization:

$$[J_1, J_2] = i\hbar J_3\,, \qquad [J_2, J_3] = i\hbar J_1\,, \qquad [J_3, J_1] = i\hbar J_2\,,$$

where $J_3 = \alpha L$ for some numerical constant α. Calculate the value of α.

(d) Consider a wavefunction of the form

$$\psi_n = NR^n e^{in\phi}\psi_0\,,$$

where n is a non-negative integer and N is a normalization constant that you do not need to determine. Check that this wavefunction is a simultaneous eigenfunction of H and L and compute the corresponding eigenvalues.

(e) Use part (d) and your knowledge of angular momentum quantization to decide whether the existence of the three conserved quantities J_1, J_2, and J_3 fully explains the degeneracies you calculated in part (a).

(f) Suppose one adds a perturbation $H' = \lambda(J_1 + 2J_3)$ with λ small. Use first order perturbation theory to determine the energy corrections to the first excited level.

3. **Square well with buckling floor**

Consider a particle of mass m in an infinite potential well with a"buckling floor". The potential is given by

$$V(x) = -V_0 \frac{(a - |x|)}{a} \quad \text{for} \quad -a < x < a$$

and $V \to \infty$ for $|x| > a$. You can assume that V_0 is small, $V_0 \ll \hbar^2/ma^2$.

a) Calculate the energies $E(n)$ of the bound states in the potential using first-order perturbation theory.

b) Calculate the energies $E(n)$ of the bound states in the potential using the WKB approximation. Leave the result in terms of an implicit equation for $E(n)$.

c) Which approximation do you expect to be more accurate for the ground state? Explain why.

Section B. Statistical Mechanics and Thermodynamics

1. **Surface Adsorption**

   Consider a 3-dimensional gas of spinless, non-relativistic, non-interacting bosons of mass m at pressure P and temperature T. The pressure of this 3D ideal Bose gas is low enough so that it is in the classical limit where the quantum statistics of the bosons may be neglected. The bosons can be adsorbed onto a 2-dimensional surface layer, where they are bound with energy $-\epsilon_0 < 0$, but retain their translational degrees of freedom in 2 dimensions. The ideal 3D Bose gas is in equilibrium with the ideal 2D adsorbed Bose gas. Treating the 2D adsorbed gas fully quantum mechanically with the proper Bose statistics, compute the surface density of this 2D gas as a function of the given parameters and fundamental constants.

   (You may need: $\int \frac{dx}{ae^x+1} = \ln \frac{e^x}{1+ae^x}$.)

2. **Polymer spring**

In a simplified model a folded polymer consists of a chain of N links ($N \gg 1$), each of length l. Each link can be in one of two possible states, forward and backward. The total length of the polymer is $L = l(N_F - N_B)$. The polymer is at a finite temperature T.

(a) Calculate the entropy of the system.

(b) Find the force with which the polymer pulls in as a function of L.

(c) In the limit $L \ll Nl$ show that the force obeys Hooke's law and calculate the spring constant.

(d) Based on your results, explain what happens when a rubber band is suddenly stretched. Will it get hotter or colder?

3. **Bernoulli's equation for gas**

Bernoulli's equation is usually discussed in relation to incompressible liquid flow, but the same principle of energy conservation along a streamline can be applied to the flow of an ideal compressible gas.

(a) Find a relationship between mass density ρ and pressure p along a streamline for an adiabatic flow of an ideal gas with $\gamma = C_p/C_V$.

(b) The flow along a stremline can be parametrized by velocity v and pressure p. Write a differential equation describing energy conservation along the streamline. Integrate this equation to obtain Bernoulli's equation for compressible flow. You can ignore the effects of gravity.

(c) Consider the gas with molar mass M escaping from a small hole in a compressed gas bottle. Inside the bottle the pressure and temperature are p_1 and T_1. Outside the bottle the atmospheric pressure is p_2. The area of the hole is much smaller than the cross-sectional area of the bottle. Find the velocity of the escaping gas.

After the exam you can check that for some ratio of pressures the escape velocity can exceed the speed of sound. In that regime, called "choked flo", the exhaust pressure is no longer equal to p_2 and the escape velocity remains equal to the speed of sound.